Clara Mendes

Embedded Systems & IoT Firmware Engineer

Specialist in Bare-Metal C, FreeRTOS, ARM Cortex-M & Ultra-Low-Power BLE

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PROFILE

Embedded firmware engineer with 6+ years experience in bare-metal C, real-time operating systems (FreeRTOS), and ultra-low-power wireless sensor networks. Specializes in hardware bring-up, microcontroller peripheral drivers, and low-energy communication protocols for industrial IoT.

EXPERIENCE

Senior Embedded Engineer 02/2021 - Present

SensorLogic IoT - Porto

Developed ultra-low-power FreeRTOS firmware for wireless industrial condition monitoring sensors on ARM Cortex-M4. Achieved 5-year continuous operation on a single coin-cell battery using deep sleep modes.

Stack: Bare-Metal C, FreeRTOS, ARM Cortex-M, STM32, BLE 5.2, Low-Power

IoT Firmware Developer 09/2018 - 01/2021

SmartGrid Devices - Aveiro

Implemented secure boot, encrypted over-the-air (OTA) firmware update protocols, and robust CAN-to-BLE gateway for EV charging stations on ESP32.

Stack: C, C++, ESP32, FreeRTOS, CAN Bus, SPI, UART, OTA Updates

Junior Hardware / Firmware Engineer 07/2017 - 08/2018

InnoWave Embedded - Porto

Designed PCB schematics, performed hardware bring-up using oscilloscopes and logic analyzers, and wrote peripheral drivers for I2C and ADC sensors.

Stack: C, KiCAD, Oscilloscopes, Logic Analyzers, I2C, SPI

Embedded Lab Assistant 09/2015 - 06/2017

INESC TEC - Porto

Programmed digital signal processing filters on TI MSP430 microcontrollers for acoustic wildlife monitoring sensor buoys.

Stack: Assembly, C, MSP430, DSP, Soldering, Hardware Testing

CORE SKILLS

Firmware & RTOS

[Bare-Metal C] [Embedded C++] [FreeRTOS] [Zephyr RTOS] [Interrupts (NVIC)] [DMA] [RTOS Mutexes]

Hardware & Microcontrollers

[ARM Cortex-M] [STM32] [ESP32] [nRF52] [I2C/SPI/UART] [CAN Bus] [ADC/DAC]

Wireless & Power Optimization

[BLE 5.2] [Low-Power Modes] [Energy Harvesting] [Zigbee] [LoRaWAN] [OTA Bootloaders]

Lab Equipment & Tooling

[Oscilloscopes] [Logic Analyzers] [JTAG/SWD] [J-Link] [KiCad] [GCC ARM] [Makefiles]

SYSTEMS ARCHITECTURE & PROJECTS

ble-sensor-beacon

Sub-10uA advertising BLE sensor firmware on nRF52840 with dynamic transmission power scaling.

can-gateway-freertos

Dual-CAN to Ethernet industrial gateway with FreeRTOS zero-copy packet queuing.

EDUCATION

B.Sc. Electrical & Computers Engineering

University of Porto (FEUP)

LANGUAGES

Portuguese	Native
English	Fluent / C2
Spanish	Working